$\boxed{\text{limit}}$

LHL

$$\left(\lim_{x \to a^-} F(x) = \lim_{h \to 0} F(a-h) \right)$$

RHL

$$\lim_{x \to a^+} F(x) = \lim_{h \to 0} F(a+h)$$

For existence of limit at $x = a$ $\Rightarrow$

$$\boxed{\lim_{x \to a^+} F(x) = \lim_{x \to a^-} F(x) = \text{Finite}}$$

24 00:03 | +4 -1

AIEEE 2011 MCQ Single Answer

+2019, 2023

$$\lim_{x\to 2}\left(\frac{\sqrt{1-\cos\{2(x-2)\}}}{x-2}\right)$$

A. Equals $\sqrt{2}$

B. Equals $-\sqrt{2}$

C. Equals $\frac{1}{\sqrt{2}}$

D. does not exist

$\sqrt{2}\left|\Delta(x-2)\right|$

$$\lim_{x \to 2} \frac{\sqrt{2}\,|\sin(x-2)|}{x-2}$$

LHL;

$$-\sqrt{2}\,\frac{\sin(x-2)}{x-2}$$

$$= -\sqrt{2}$$

RHL;

$$\sqrt{2}\,\frac{\sin(x-2)}{x-2}$$

$$= \sqrt{2}$$

Indeterminate Forms

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad \infty - \infty, \quad 0 \times \infty, \quad 1^{\infty}, \quad \infty^{0}, \quad 0^{0}$$

→ tends to 0

→ tends to 1

$$\boxed{0^{\infty} = 0}$$

→ All forms reduced to 0/0 form.

$$y = 1^{\infty} \Rightarrow \log y = \infty \times \log(1) = 0 \times \infty = \frac{0}{1/\infty}$$

Algebra of limits

$$\lim_{x \to a} f(x) = k \qquad \lim_{x \to a} g(x) = s$$

$\boxed{k, s \text{ exist}}$

$$\boxed{\begin{gathered}\lim_{x \to a} (f \pm g)(x) \\ = \\ \lim f(x) + \lim g(x) \\ k + s\end{gathered}}$$

$$\lim_{x \to a} f(x) \cdot g(x) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x) = ks$$

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{k}{s} \qquad (s \neq 0)$$

- $$\lim_{x \to a} n\, F(x) = n \cdot \lim_{x \to a} F(x)$$

$$\lim_{x \to a} F(x)^{g(x)} = \left(\lim_{x \to a} F(x) \right)^{\lim_{x \to a} g(x)} = K^{\Delta}$$

$$\boxed{\lim_{x \to a} F \circ g(x) = F\left(\lim_{x \to a} g(x) \right) = F(\Delta)}$$

if F is continuous

at $g(x) = \Delta$

exi

$$\lim_{x \to a} \log F(x) = \log\left(\lim_{x \to a} F(x)\right) = \log(k)$$

$$\lim_{x \to a} e^{F(x)} = e^{\lim_{x \to a} F(x)} = e^{k}.$$

Standard limits

$$\lim_{x\to 0}\left(\frac{\sin(x)}{x}, \frac{x}{\sin(x)}, \frac{\sin^{-1}(x)}{x}, \frac{x}{\sin^{-1}(x)}\right) = 1$$

$$\lim_{x\to 0}\left(\frac{\tan(x)}{x}, \frac{x}{\tan(x)}, \frac{\tan^{-1}(x)}{x}, \frac{x}{\tan^{-1}(x)}\right) = 1$$

$$\boxed{\lim_{x\to 0} \frac{1-\cos(x)}{x^2} = \frac{1}{2}}$$

73 01:05 | +4 -1

JEE Main 2019 (Online) 11th January Evening Slot | MCQ Single Answer

$\lim\limits_{x \to 0} \frac{x\cot(4x)}{\sin^2 x \cot^2(2x)}$ is equal to :

(A) 0

(B) 4

(C) 1

(D) 2

$$\cancel{x}\cdot\frac{4x}{\boxed{t(4x)}} \times \underbrace{\frac{x^2}{s^2(x)}} \times \frac{t^2(2x)}{(2x)^2}$$

$$\cancel{x}\,\frac{1}{\cancel{4x}} \times \frac{1}{\cancel{x^2}} \times (\cancel{2x})^2 \quad \cancel{4}$$

77

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JEE Main 2020 (Online) 3rd September Morning Slot

Numerical

If $\lim_{x \to 0} \left\{ \frac{1}{x^8} \left(1 - \cos\frac{x^2}{2} - \cos\frac{x^2}{4} + \cos\frac{x^2}{2}\cos\frac{x^2}{4} \right) \right\} = 2^{-k}$

then the value of k is ______ .

$$\left(\frac{x^2}{2}\right)^2 \times \left(\frac{x^2}{4}\right)^2 \times \frac{1}{x^8} \quad \frac{\left(1 - c\left(\frac{x^2}{2}\right)\right)}{\left(\frac{x^2}{2}\right)^2} \quad \frac{\left(1 - c\left(\frac{x^2}{4}\right)\right)}{\left(\frac{x^2}{4}\right)^2}$$

$$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4^2} = 2^{-8}$$

H.W

30 00:08 | +4 -1

JEE Main 2015 (Offline) MCQ Single Answer

$\lim_{x \to 0} \frac{(1-\cos 2x)(3+\cos x)}{x \tan 4x}$ is equal to

A 2

B $\frac{1}{2}$

C 4

D 3

$$\boxed{\lim_{x \to 0} \frac{\sin(\pi \cos^2(x))}{x^2}}$$

∴ Hint

$$\boxed{\sin(\pi - \theta) = \sin(\theta)}$$

- $$\lim_{x \to 0} \frac{a^x - 1}{x} = \ln(a)$$

- $$\lim_{x \to 0} \frac{e^x - 1}{x} = \ln(e) = 1$$

- $$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1$$

- $$\lim_{x \to a} \frac{x^n - a^n}{x - a} = \frac{n \cdot a^{n-1}}{}$$

85 00:03 | +4 -1

JEE Main 2020 (Online) 5th September Evening Slot

MCQ Single Answer

$$\lim_{x \to 0} \frac{x\left(e^{\left(\sqrt{1+x^2+x^4}-1\right)/x}-1\right)}{\sqrt{1+x^2+x^4}-1}$$

A is equal to 0.

B is equal to $\sqrt{e}$.

C is equal to 1.

D does not exist.

$$\Rightarrow \lim_{p \to 0} \frac{e^p - 1}{p} = 1$$

???

$$p = \frac{\sqrt{1+x^2+x^4} - 1}{x}$$

$$\lim_{x\to 0} \frac{x\left(e^{\left(\sqrt{1+x^2+x^4}-1\right)/x}-1\right)}{\sqrt{1+x^2+x^4}-1}$$

$$= \lim_{x\to 0} \frac{x\left[e^{\frac{\left(\sqrt{1+x^2+x^4}-1\right)\left(\sqrt{1+x^2+x^4}+1\right)}{x\left(\sqrt{1+x^2+x^4}+1\right)}}-1\right]\times\left(\sqrt{1+x^2+x^4}+1\right)}{\left(\sqrt{1+x^2+x^4}-1\right)\left(\sqrt{1+x^2+x^4}+1\right)}$$

$$= \lim_{x\to 0} \frac{x\left[e^{\frac{\left(1+x^2+x^4\right)-1}{x\left(\sqrt{1+x^2+x^4}+1\right)}}-1\right]\times\left(\sqrt{1+x^2+x^4}+1\right)}{(1+x^2+x^4-1)}$$

$$= \lim_{x\to 0} \frac{\left[e^{\frac{x^2+x^4}{x\left(\sqrt{1+0+0}+1\right)}}-1\right]\times\left(\sqrt{1+0+0}+1\right)}{x+x^3}$$

$$= 2\lim_{x\to 0} \frac{\left[e^{\frac{x^2+x^4}{2x}}-1\right]}{x+x^3}$$

$$= 2\lim_{x\to 0} \frac{\left[e^{\frac{x+x^3}{2}}-1\right]}{\frac{x+x^3}{2}\times 2}$$

$$a^{\infty} = 0 \quad \text{if} \quad a \in [0, 1]$$

67

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JEE Main 2023 (Online) 24th January Morning Shift

MCQ Single Answer

$\lim\limits_{t \to 0}\left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \ldots + n^{\frac{1}{\sin^2 t}}\right)^{\sin^2 t}$ is equal to

$\boxed{\infty^0}$

A $\frac{n(n+1)}{2}$

B n

C $n^2 + n$

D n^2

$$\left(n^{1/s^2(t)}\right)^{s^2(t)} \left[\left(\frac{1}{n}\right)^{\operatorname{cosec}^2(t)} + \left(\frac{2}{n}\right)^{\operatorname{cosec}^2(t)} + \cdots + 1 \right]^{s^2(t)}$$

$$n\left[0 + 0 + \cdots + 1 \right]^{0}$$

$$= n$$

$$\lim_{x \to 0} \frac{\sin(x) - x}{x^3}$$

$$= -\frac{1}{6}$$

$$\lim_{x \to 0} \frac{\tan(x) - x}{x^3}$$

$$= \frac{1}{3}$$

Solving Forms

$\lim_{x \to a} \frac{F(x)}{g(x)}$ takes $\frac{0}{0}$ or $\frac{\infty}{\infty}$ Form

then equals to $\boxed{\lim_{x \to a} \frac{F'(x)}{g'(x)}}$

till indet. Form.

Form

$0^0, \infty^0$

$$\lim_{x \to a} \left(F(x)\right)^{g(x)} = L$$

$$\ln(L) = \lim_{x \to a} g(x) \cdot \ln\left(F(x)\right)$$

$$\boxed{L = e^{\lim_{x \to a} g(x) \cdot \ln\left(F(x)\right)}}$$

$$\boxed{1^{\infty}}$$

$$L = \lim_{x \to a} \left(F(x) \right)^{g(x)}$$

if $\lim_{x \to a} F(x) = 1$

and

$\lim_{x \to a} g(x) = \infty$

$$\boxed{L = e^{\lim_{x \to a} \left(F(x) - 1 \right) \cdot g(x)}}$$

1

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JEE Main 2025 (Online) 3rd April Evening Shift

Numerical

If $\lim_{x \to 0} \left(\frac{\tan x}{x}\right)^{\frac{1}{x^2}} = p$, then $96 \log_e p$ is equal to______ 32

Enter Answer

$$e^{\lim_{x \to 0} \frac{1}{x^2}\left(\frac{\tan(x)}{x} - 1\right)} = e^{1/3}$$

$$\frac{\tan(x) - x}{x^3}$$

45

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IIT-JEE 2011 Paper 2 Offline

MCQ Single Answer

If $\lim\limits_{x \to 0} [1 + x \ln(1 + b^2)]^{1/x} = 2b\sin^2\theta, b > 0$ and $\theta \in (-\pi, \pi]$, then the value of θ is

A $\pm\frac{\pi}{4}$

B $\pm\frac{\pi}{3}$

C $\pm\frac{\pi}{6}$

D $\pm\frac{\pi}{2}$

$$\frac{1+b^2}{2b} = \sin^2(\theta)$$

$$\left[\frac{1}{b} + b\right] = 2\Delta^2(\theta)$$

Expansions

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - - - - - -$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - - - - - -$$

$$\tan(x) = x + \frac{x^3}{3} + \frac{2}{15}x^5 + - - - - - -$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + - - - - - -$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4}$$

$$\boxed{|x| \leq 1}$$

$$\sin^{-1}(x) = x + \frac{x^3}{6} + \frac{3x^5}{40} + \cdots$$

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\cdots$$

$$(1+x)^{1/x} = e\left[1 - \frac{x}{2} + \frac{11}{24}x^2 \cdots\cdots\right]$$

137

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JEE Main 2021 (Online) 18th March Morning Shift

MCQ Single Answer

If $\lim\limits_{x \to 0} \frac{\sin^{-1} x - \tan^{-1} x}{3x^3}$ is equal to L, then the value of (6L + 1) is

A $\frac{1}{6}$

B $\frac{1}{2}$

C 6

D 2

$$\lim_{x\to 0} \frac{\left(x+\frac{x^3}{6}---\right)-\left(x-\frac{x^3}{3}\right)}{3x^3}$$

$$\frac{\cancel{x^3}\left(\frac{1}{6}+\frac{1}{3}\right)}{\cancel{x^3}\,3}$$

47

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JEE Main 2024 (Online) 27th January Evening Shift

MCQ Single Answer

If $\lim\limits_{x \to 0} \frac{3+\alpha \sin x+\beta \cos x+\log_e(1-x)}{3\tan^2 x} = \frac{1}{3}$, then $2\alpha - \beta$ is equal to :

x^2

x^2

1

(A) 2

(B) 1

(C) 5

(D) 7

$$\frac{3 + \alpha\left(x - \frac{x^3}{3!}\right) + \beta\left(1 - \frac{x^2}{2!}\right) + \left(-x - \frac{x^2}{2}\right)}{3x^2}$$

$$\frac{(\beta+3) + (\alpha - 1)x + \frac{\beta}{2}\left(-x^2\right)}{3x^2}$$

40

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JEE Main 2024 (Online) 9th April Evening Shift

MCQ Single Answer

$\lim_{x \to 0} \frac{e-(1+2x)^{\frac{1}{2x}}}{x}$ is equal to

(A) $\frac{-2}{e}$

(B) $e - e^2$

(C) 0

(D) e

$$\frac{\cancel{e} - (\cancel{e} - x)}{x} = \frac{\cancel{x}}{\cancel{x}} = 1$$

$$(1+x)^{1/x} = e\left(1 - \frac{x}{2} + \frac{11}{24}x^2 \cdots\cdots\right)$$

$$= e\left(1 - x + 1\right)$$

Sandwich theorem

$$F(x) \leq g(x) \leq h(x) \qquad \forall\ x \in (\alpha, \beta) - \{a\}$$

$$\lim_{x \to a} F(x) = \lim_{x \to a} h(x) = L$$

$$\boxed{\lim_{x \to a} g(x) = L}$$

For $a \in (\alpha, \beta)$.

48 00:02 | +4 -1

JEE Main 2018 (Offline) MCQ Single Answer

For each $t \in R$, let [t] be the greatest integer less than or equal to t.

Then $\lim_{x \to 0^+} x\left(\left[\frac{1}{x}\right]+\left[\frac{2}{x}\right]+\ldots\ldots+\left[\frac{15}{x}\right]\right)$

(A) does not exist in R

(B) is equal to 0

(C) is equal to 15

(D) is equal to 120

$[x] = x - \{x\}$

$x-1 < [x] \leq x$

$$x\left[\left(\frac{1}{x}+\frac{2}{x}+\cdots\frac{15}{x}\right)-\left(\left\{\frac{1}{x}\right\}+\left\{\frac{2}{x}\right\}\cdots\left\{\frac{15}{x}\right\}\right)\right]$$

$$(1+\cdots\cdots15)-x(\qquad)$$

$$\frac{1}{x} - 1 \le \left[\frac{1}{x}\right] \le \frac{1}{x}$$

$$\frac{2}{x} - 1 \le \left[\frac{2}{x}\right] \le \frac{2}{x}$$

$$\vdots$$

$$\frac{15}{x} - 1 \le \left[\frac{15}{x}\right] \le \frac{15}{x}$$

$$\Rightarrow x\left(\frac{1}{x}+\frac{2}{x}+\cdots\frac{15}{x}\right)-15x \le \boxed{\left(\left\lceil\frac{1}{x}\right\rceil+\left\lceil\frac{2}{x}\right\rceil+\cdots\left\lceil\frac{15}{x}\right\rceil\right)\times x}$$

$$\le \left(\frac{1}{x}+\frac{2}{x}+\cdots\frac{15}{x}\right)\times x$$

$$0$$

$$120-15x \le \boxed{0} \le (1+\cdots 15)$$